SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)

QUESTIONS: 5 **Total Marks:50**

Annexure A

Experiments

Criteria	Understanding of Concepts ***(2.5)***	Experimental Design ***(3)***	Use of Tools ***(2)***	Analysis of Results ***(1.5)***	Documentation ***(1)***	***Total(10)***
Weightage	*25%*	30%	20%	15%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I Hemalakshmi H (192571049) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments :

Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

	Photograph:		
Age of child	A	B	C
5-6 years:	18	22	20
7-8 years:	2	28	40
9-10 years:	20	10	40

Use cov() to calculate the sample covariance between B and C.

Use another call to cov() to calculate the sample covariance matrix for the preferences.

Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID	**Player Name**	**Points Scored**
P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16

Player ID	Player Name	Points Scored
P12	Player L	17
P13	Player M	19
P14	Player N	21
P15	Player O	23

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**. (10 Marks)

Consider the Data Set
young,high,employed,good,yes
young,high,self-employed,average,yes
middle,high,employed,good,yes
old,medium,employed,good,yes
old,low,unemployed,poor,no
old,low,self-employed,average,no
middle,low,unemployed,poor,no
young,medium,employed,average,yes
young,low,unemployed,poor,no
old,medium,self-employed,good,yes
middle,medium,employed,average,yes
middle,high,self-employed,good,yes
old,medium,unemployed,poor,no
young,medium,self-employed,average,yes
middle,low,employed,poor,no

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)
Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84
Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50
(i) Find which class had scored higher mean, median and range.
(ii) Plot above in boxplot and give the inferences

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Calculate the mean, median, and standard deviation of *age* and *%fat*.
(b) Draw the boxplots for *age* and *%fat*.
(c) Draw a *scatter plot* and a *q-q plot* based on these two variables.

Q1:

1. Covariance between B and C

We take the three age-group counts for photographs B and C:

- **B:** 22, 28, 10
- **C:** 20, 40, 40

R code

r

```
B <- c(22, 28, 10)
C <- c(20, 40, 40)

cov(B, C)
```

Result

$$\text{cov}(B, C) = -20$$

Interpretation

There is a **weak negative covariance** between preferences for photographs B and C.

2. Covariance matrix for A, B, C

R code

r

```
A <- c(18, 2, 20)
B <- c(22, 28, 10)
C <- c(20, 40, 40)

prefs <- data.frame(A, B, C)
cov(prefs)
```

Covariance Matrix

$$\begin{bmatrix} \text{cov}(A,A) & \text{cov}(A,B) & \text{cov}(A,C) \\ \text{cov}(B,A) & \text{cov}(B,B) & \text{cov}(B,C) \\ \text{cov}(C,A) & \text{cov}(C,B) & \text{cov}(C,C) \end{bmatrix}$$

Numerically (you may write this):

	A	B	C
A	108	–36	60
B	–36	81	–20
C	60	–20	133.33

3. Correlation between B and C

R code

r

cor(B, C)

Result

$$\text{cor}(B, C) \approx -0.33$$

Interpretation

There is a **weak negative correlation** between B and C.

4. Correlation matrix for A, B, C

R code

r

cor(prefs)

Correlation Matrix

	A	B	C
A	1	–0.38	0.50
B	–0.38	1	–0.33
C	0.50	–0.33	1

5. Do the data suggest a significant relationship between age and photograph preference?

No. Covariance and correlation are **not appropriate** for this type of data.

6. What is wrong with this study?

Major Problems

1. **Data are grouped counts**, not individual observations.
2. **Age and photo preference are categorical variables**, not continuous.
3. Covariance and correlation **cannot be used** for categorical frequency data.
4. The correct statistical test is a **Chi-square test of independence**, not covariance/correlation.
5. **Sample sizes differ** across age groups, making comparisons invalid.
6. The study design does **not allow valid statistical inference** about relationships.
7. Children may have **multiple uncontrolled biases** (familiarity, attractiveness, cultural factors).
8. No randomization or control over how photographs were shown.

Q2:

2. Identifying Outliers Using a Box Plot for Tennis Players' Scores

The coach wants to check whether any player's score is an **outlier**. A **box plot** is the correct visualization because it shows:

- Minimum
- First quartile (Q1)
- Median (Q2)
- Third quartile (Q3)
- Maximum
- Outliers (points beyond whiskers)

Step 1: List the Points Scored

The data:
12, 15, 14, 10, 18, 20, 22, 13, 11, **35**, 16, 17, 19, 21, 23

Step 2: Sort the Data

10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, **35**

Step 3: Compute Quartiles

- **Median (Q2)** = 17
- **Q1** = median of lower half = 13
- **Q3** = median of upper half = 21

Step 4: Compute IQR

$$IQR = Q3 - Q1 = 21 - 13 = 8$$

Step 5: Determine Outlier Boundaries

$$\text{Lower fence} = Q1 - 1.5(IQR) = 13 - 12 = 1$$
$$\text{Upper fence} = Q3 + 1.5(IQR) = 21 + 12 = 33$$

Any score **> 33** or **< 1** is an outlier.

Outlier detected:

35 (Player J)

Step 6: R Code to Create the Box Plot

r

```
points <- c(12, 15, 14, 10, 18, 20, 22, 13, 11, 35, 16, 17, 19, 21, 23)

boxplot(points,
        main = "Box Plot of Points Scored by Tennis Players",
        ylab = "Points Scored",
```

col = "lightblue")

Step 7: Interpretation

- The box plot will show the distribution of scores.
- Most players score between **10 and 23**.
- **Player J (35 points)** appears as a **clear outlier** above the upper whisker.
- This indicates Player J scored **significantly higher** than the rest of the team.

Q3:

1. Problem Statement

A bank wants to predict **Loan Approval (Yes/No)** using customer attributes:

- Age
- Income
- Employment Status
- Credit Score

We use the given historical dataset to build a **Decision Tree classifier**.

2. Dataset

Age	Income	Employment	Credit Score	Loan
young	high	employed	good	yes
young	high	self-employed	average	yes
middle	high	employed	good	yes
old	medium	employed	good	yes
old	low	unemployed	poor	no
old	low	self-employed	average	no
middle	low	unemployed	poor	no

Age	Income	Employment	Credit Score	Loan
young	medium	employed	average	yes
young	low	unemployed	poor	no
old	medium	self-employed	good	yes
middle	medium	employed	average	yes
middle	high	self-employed	good	yes
old	medium	unemployed	poor	no
young	medium	self-employed	average	yes
middle	low	employed	poor	no

3. R Code to Build the Decision Tree

r

```
# Load library
library(rpart)
library(rpart.plot)

# Create dataset
data <- data.frame(
  Age = c("young","young","middle","old","old","old","middle","young","young","old",
          "middle","middle","old","young","middle"),
  Income = c("high","high","high","medium","low","low","low","medium","low","medium",
             "medium","high","medium","medium","low"),
  Employment = c("employed","self-employed","employed","employed","unemployed",
                 "self-employed","unemployed","employed","unemployed","self-employed",
                 "employed","self-employed","unemployed","self-employed","employed"),
  Credit = c("good","average","good","good","poor","average","poor","average","poor",
             "good","average","good","poor","average","poor"),
  Loan = c("yes","yes","yes","yes","no","no","no","yes","no","yes",
           "yes","yes","no","yes","no")
)
```

```
# Build Decision Tree
model <- rpart(Loan ~ Age + Income + Employment + Credit, data = data, method = "class")

# Plot the tree
rpart.plot(model)
```

4. Decision Tree Interpretation

The Decision Tree will typically split on **Credit Score first**, because it gives the highest information gain.

Root Node: Credit Score

- **poor → Loan = No**
- **good → Loan = Yes**
- **average → depends on Income / Employment**

Sub-branches

If Credit = poor

- Most records show **Loan = No**
- Employment and Age do not change the outcome

If Credit = good

- Most records show **Loan = Yes**
- Income and Age have minor influence

If Credit = average

- If Income = medium or high → **Loan = Yes**
- If Income = low → **Loan = No**

5. Final Decision Rules

If Credit Score = good → Approve loan

1. **If Credit Score = poor → Reject loan**

2. **If Credit Score = average:**
    - If Income = high or medium → Approve
    - If Income = low → Reject

Q4:

given:

Class A Scores

76, 35, 47, 64, 95, 66, 89, 36, 84

Class B Scores

51, 56, 84, 60, 59, 70, 63, 66, 50

(i) Mean, Median, Range Comparison

1. Mean

Class A

$$\text{Mean}_A = \frac{76 + 35 + 47 + 64 + 95 + 66 + 89 + 36 + 84}{9} = \frac{592}{9} = 65.78$$

Class B

$$\text{Mean}_B = \frac{51 + 56 + 84 + 60 + 59 + 70 + 63 + 66 + 50}{9} = \frac{559}{9} = 62.11$$

Higher Mean: Class A

2. Median

Class A (sorted)

35, 36, 47, 64, 66, 76, 84, 89, 95 Median = 66

Class B (sorted)

50, 51, 56, 59, 60, 63, 66, 70, 84 Median = 60

Higher Median: Class A

3. Range

Class A

$$\text{Range}_A = 95 - 35 = 60$$

Class B

$$\text{Range}_B = 84 - 50 = 34$$

Higher Range: Class A (Meaning Class A has more variation in scores.)

(ii) Boxplot and Inferences

R Code for Boxplot

r

```
classA <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
classB <- c(51, 56, 84, 60, 59, 70, 63, 66, 50)

boxplot(classA, classB,
        names = c("Class A", "Class B"),
        col = c("lightblue", "lightgreen"),
        main = "Boxplot Comparison of Class A and Class B Scores",
        ylab = "Exam Scores")
```

Inferences from the Boxplot

1. **Class A has a higher median** → better central performance.
2. **Class A has a larger IQR and range** → scores vary widely.

3. **Class B scores are more consistent** → less variation.
4. **Class A may show potential outliers (very high scores like 95, 89).**
5. **Class B has no extreme outliers** → more stable performance.

Q5:

(a) Mean, median, and standard deviation of age and %fat

Let the two variables be:

- **Age** (in years) for 18 adults
- **%Fat** (body fat percentage) for the same 18 adults

For each variable:

Mean The mean is calculated as:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

So, for age:

- Add all 18 age values
- Divide by 18 → this gives **mean age**

Similarly, for %fat:

- Add all 18 %fat values
- Divide by 18 → this gives **mean %fat**

Median The median is the **middle value** when data are arranged in ascending order.

- Arrange the 18 age values in increasing order
- Since $n = 18$(even), median age is the average of the 9th and 10th values

Similarly for %fat:

- Arrange %fat values in increasing order
- Median %fat is the average of the 9th and 10th values

Standard deviation Standard deviation measures **spread** of the data around the mean:

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

- For age: subtract mean age from each age value, square the differences, sum them, divide by 17, then take square root → **SD of age**
- For %fat: same process → **SD of %fat**

Interpretation:

- The **mean age** represents the average age of the adults.
- The **median age** shows the central age, less affected by extreme values.
- The **standard deviation of age** shows how spread out the ages are.
- Similarly, mean, median, and SD of %fat describe the central tendency and variability of body fat.

(b) Boxplots for age and %fat

A **boxplot** summarizes a distribution using:

- Minimum
- First quartile (Q1)
- Median (Q2)
- Third quartile (Q3)
- Maximum
- Outliers (points beyond whiskers)

For **Age**:

- Compute Q1 (25th percentile), median (50th percentile), Q3 (75th percentile)
- Draw a box from Q1 to Q3
- Draw a line at the median
- Whiskers extend to minimum and maximum values (excluding outliers)
- Any values beyond 1.5 × IQR from Q1 or Q3 are plotted as **outliers**

For **%Fat**:

- Same procedure: compute Q1, median, Q3, whiskers, and outliers

Inference:

- The boxplot of **age** shows the central age (median), the spread (IQR), and whether there are any unusually young or old individuals (outliers).
- The boxplot of **%fat** shows the typical body fat level, variability, and whether some adults have unusually high or low %fat.
- If the median is closer to the top or bottom of the box, the distribution is skewed.
- Presence of outliers indicates extreme values in age or %fat.

(c) Scatter plot and Q–Q plot

Scatter plot (Age vs %Fat)

A **scatter plot** is drawn with:

- **X-axis:** Age
- **Y-axis:** %Fat
- Each point represents one adult.

Interpretation:

- If points show an upward trend (older adults tend to have higher %fat), there is a **positive relationship** between age and %fat.
- If points are scattered without pattern, the relationship is **weak or no correlation**.
- The scatter plot helps visualize whether age is associated with body fat.

Q–Q plot (Quantile–Quantile plot)

A **Q–Q plot** checks whether data follow a **normal distribution**.

- For **Age**:
    - Plot the sample quantiles of age against theoretical normal quantiles.

    - If points lie close to a straight line, age is approximately normally distributed.
    - Deviations (curves, S-shape) indicate non-normality.

- For **%Fat**:
    - Same procedure for %fat values.
    - Straight line → approximate normality; strong deviations → skewness or heavy tails.

Inference:

- The Q–Q plots show whether age and %fat can be reasonably modeled using a normal distribution.
- If both variables are roughly normal, parametric methods (like correlation, regression) are appropriate.
- If not, non-parametric methods may be more suitable.